Reactivity of metals

Some metals are more reactive than others. Most (active) metals can be made to react with oxygen, water, acids and alkalis. Metals such as gold (Au) and silver (Ag) are very unreactive.

Metal + Oxygen → Metal oxide

1 Write a balanced formula equation for each reaction.

a Copper + oxygen → copper (II) oxide

b Iron + oxygen → iron (III) oxide

Metal + Water → Metal hydroxide + Hydrogen

2 Write a balanced formula equation for each reaction.

a Sodium + water → sodium hydroxide + hydrogen

b Potassium + water → potassium hydroxide + hydrogen

c Calcium + water → calcium hydroxide + hydrogen

Metal + Acid → Salt + Hydrogen

Metals such as magnesium (Mg), zinc (Zn) and aluminium (Al) react with acids.

3 Write a balanced formula equation for each reaction.

a Zinc + hydrochloric acid → zinc chloride + hydrogen

b Magnesium + hydrochloric acid → magnesium chloride + hydrogen

c Magnesium + sulfuric acid → magnesium sulfate + hydrogen

Recycling metals

Our supply of metals or minerals is limited by how much of them is in the ground and how much of that is obtainable. Since 1950, more metals have been used up than in the whole history of the world before 1950. It is difficult to estimate how long our supply of metals will last, so we should be more careful about how much metal we waste. As metals become more expensive, recycling becomes more attractive. Producing copper from scrap costs about 4% of the cost of making it from its ore. Today about 50% of copper and about 80% of the lead from car batteries is recycled. About 20% of aluminium and 18% of iron (as steel) is recycled.

4 Use the example of aluminium drink cans to outline the steps in the process of recycling aluminium. Discuss whether recycling aluminium requires the same amount of energy as extracting it from ore.

5 Outline reasons, other than cost, why containers should be recycled.

6 Outline why aluminium is a good material for making drink containers.

Metal alloys

Metal alloys are mixtures of two or more metals. They take advantage of the properties of a number of metals in order to produce properties that are superior to those of pure metals. Bronze is an alloy of copper and tin. It is hard and does not corrode. Cupronickel is 75% copper and 25% nickel. It is an attractive silver in colour and is hardwearing. Steel is an alloy of iron with some carbon added. There any many types of steel and all of these contain iron, but the other metals present vary. Brass contains roughly 70% copper and 30% zinc. It dulls in appearance but doesn't corrode. Duraluminium contains 94.5% aluminium, 4% copper, 1% manganese and 0.05% magnesium. This alloy is three times as strong as steel, but only one-third as dense. Solder is 70% tin and 30% lead. It is soft and has a low melting point.

7 Why are alloys used instead of pure metals?

8 Draw up a table to present the information on alloys above. Include the metals that make up alloys, their properties and possible uses.

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